

SCHOOL OF COMPUTER SCIENCE AND ENGINEERING (SCOPE)

CBS3009 ADVANCED SOCIAL TEXT AND MEDIA ANALYTICS

PROJECT WORK

B.Tech [CSBS]

AY:2026-2027

AUTOMATED TEXT MINING DASHBOARD FOR EXTRACTING SAFETY HAZARD VIOLATIONS FROM BUILDING SITE INSPECTION REPORTS USING SPACY AND STREAMLIT

A PROJECT REPORT

Submitted by

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DECLARATION

I hereby declare that this project report entitled “Automated Text Mining Dashboard for Extracting Safety Hazard Violations from Building Site Inspection Reports using spaCy and Streamlit” has been prepared as part of CBS3009 – Advanced Social Text and Media Analytics. The report describes the project design, data preparation, NLP pipeline, classification methods, dashboard, and evaluation considerations. Student, registration, and guide details must be completed and verified before submission.

Student Signature:  Date: 26th October 2026

ACKNOWLEDGEMENT

I would like to thank Dr. Mythili T (HOD) for reviewing the entire project and guide us for the team leader and also helped me to coordinate my project. After completion of my project, I would like to take this chance to express my sincere gratitude to my project guide Dr. Kavitha G who has guided me thoroughly throughout my project development. I would like to express my gratitude to all our friends and teachers, who helped and cooperated directly or indirectly in the accomplishment of this project. Heartfelt appreciation to those who helped me by cooperative acts, logical device to beautify this project to be effective and user-friendly.

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ABSTRACT

Construction-site inspection reports contain important information about unsafe conditions, non-compliance, and incidents, but this information is often recorded as unstructured narrative text. Manual review of large collections can be time-consuming and makes consistent comparison difficult. This project presents an Automated Text Mining Dashboard for Extracting Safety Hazard Violations from Building Site Inspection Reports using spaCy and Streamlit. The system accepts PDF, DOCX, and TXT reports, extracts and cleans text, segments it into sentences, and identifies candidate hazards using a two-stage approach: curated phrase matching with spaCy PhraseMatcher and semantic similarity based on sentence embeddings. A separate supervised text-classification component predicts labels learned from a prepared dataset. Candidate models include TF-IDF with Linear Support Vector Machine (SVM), Complement Naive Bayes, BiLSTM with attention, DistilBERT, and DeBERTa-v3-small, subject to compatible local model files and dependencies. The Streamlit dashboard presents findings, evidence sentences, predicted labels, scores where available, visual summaries, and structured exports. Working data exploration included an injury-severity spreadsheet with approximately 53,550 records and 24 fields, around 10,299 construction-related records identified during an earlier filtering step, and a separate cleaned text-classification file with approximately 48,985 rows. These counts represent different data stages and must be reconciled against the final processing scripts. Preliminary results recorded for a common benchmark split show 91.65% accuracy and 77.49% macro-F1 for Linear SVM, and approximately 82.55% accuracy and 72.64% macro-F1 for Complement Naive Bayes. Severe class imbalance and rare labels limit interpretation; numeric label meanings must be verified before assigning severity names. The system is a decision-support prototype, not a replacement for qualified safety inspectors or a definitive regulatory-compliance tool.

Keywords: construction safety, text mining, spaCy, Streamlit, hazard detection, TF-IDF, Linear SVM, sentence embeddings, classification.

1. INTRODUCTION

1.1 Background

Construction sites involve work at height, temporary structures, heavy machinery, electrical installations, material handling, and changing site conditions. Inspection reports record observations, unsafe acts, corrective actions, and incident narratives. Much of this information is written in free text, making large-scale comparison and retrieval difficult. Natural language processing (NLP) can convert narrative text into structured sentence-level information that can be searched, summarized, and reviewed.

1.2 Problem Statement

Manual inspection of many reports is labour-intensive and may lead to inconsistent categorization or missed evidence. Keyword-only search can miss paraphrases, while statistical classifiers may not expose the sentence that triggered a prediction. The project combines transparent phrase matching, semantic matching, supervised classification, and an interactive dashboard that keeps supporting evidence visible.

1.3 Aim and Objectives

- Build a parser for PDF, DOCX, and TXT inspection reports.
- Clean extracted text and segment it into sentences using spaCy.
- Detect candidate hazards through curated phrases, contextual checks, and semantic similarity.
- Evaluate supervised text classifiers against prepared labels.
- Present findings, evidence, distributions, and exports in Streamlit.
- Use class-aware metrics and document limitations caused by imbalanced data.

1.4 Scope and Expected Outcome

The system highlights candidate hazards and predicts the class labels represented in the training data. It does not certify regulatory compliance, establish legal liability, or replace an inspector. The intended outcome is a reviewable dashboard that makes narrative reports easier to search, summarize, and export.

2. LITERATURE REVIEW

2.1 Construction Accident Text Mining

Zhang et al. (2019) studied construction-site accident analysis using text mining and NLP. Their work motivates extracting structured patterns from accident narratives. The present project focuses on an interactive review workflow that retains sentence-level evidence.

2.2 Injury Precursors and Supervised Learning

Baker et al. (2020) investigated automatically learning construction injury precursors from text. This is relevant to the supervised component, where models learn patterns from historical narratives. Consistent labels and sufficient examples for rare outcomes remain essential.

2.3 Hybrid Accident Classification

Cheng et al. (2020) examined hybrid supervised machine learning for construction-site accident classification. This supports comparing an efficient sparse-text baseline with neural and transformer alternatives rather than assuming one algorithm is best for every dataset.

2.4 Open Data and Inspection Information Extraction

Rupasinghe and Panuwatwanich (2021) applied text mining to open data for understanding construction hazards. Liu et al. (2023) studied graph-based information extraction from government on-site inspection reports. Together, these studies highlight the value of converting narrative safety records into structured information and documenting data provenance.

2.5 Research Gap and Project Positioning

The project integrates two hazard-detection paths: rule-based phrase matching and semantic similarity with supervised classification and a dashboard. Rule-based matching is transparent; semantic matching can handle related wording; classification assigns a learned label; and the dashboard allows reviewers to inspect evidence instead of blindly accepting model outputs. The system remains a prototype that requires validation on manually annotated reports.

Study	Focus	Relevance
Zhang et al. (2019)	Text mining/NLP for construction accidents	Narrative analysis motivation.
Baker et al. (2020)	Learning injury precursors	Supervised learning from text.
Cheng et al. (2020)	Hybrid accident classification	Comparative model evaluation.

Rupasinghe & Panuwatwanich (2021)	Open-data hazard mining	Data provenance and hazard categories.
Liu et al. (2023)	Inspection-report information extraction	Structured extraction from reports.

3. HARDWARE AND SOFTWARE REQUIREMENTS

Component	**Purpose / Requirement**
CPU	Modern multi-core processor for parsing and classical NLP.
Memory	8 GB RAM is a practical minimum; transformer inference benefits from additional memory.
Storage	Datasets, model weights, uploaded reports, and exports.
GPU	Optional; useful for transformer training or accelerated inference.
Python	Core implementation language.
Streamlit	Interactive dashboard, uploads, charts, and downloads.
spaCy	Sentence segmentation, tokenization, linguistic processing, PhraseMatcher.
PyMuPDF (fitz)	PDF text extraction.
python-docx	DOCX parsing.
pandas / NumPy	Tabular processing and exports.
scikit-learn	TF-IDF, Linear SVM, Complement Naive Bayes, metrics.
PyTorch / Transformers	Neural and transformer inference when dependencies and model assets are available.
sentence-transformers	Sentence embeddings and semantic similarity.
Plotly / Matplotlib	Visualizations.
joblib	Loading compatible serialized scikit-learn pipelines.

Exact package versions should be recorded from the final environment before deployment. A model-loading failure should display the underlying exception so that path, tokenizer, dependency, and checkpoint problems can be diagnosed.

4. METHODOLOGY AND SYSTEM DESIGN

4.1 Overall Architecture

The uploaded report is parsed into text, cleaned, and segmented into sentences. The NLP stage provides sentence boundaries and linguistic processing. Hazard detection runs rule-based and semantic paths; findings are then consolidated and deduplicated. A separate classifier predicts a learned class label for each relevant sentence. The dashboard presents results with evidence and export options.

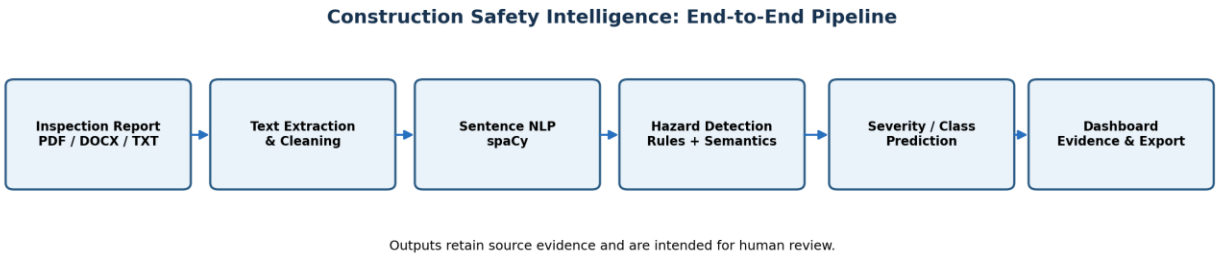


Figure 1. End-to-end construction safety text-mining pipeline.

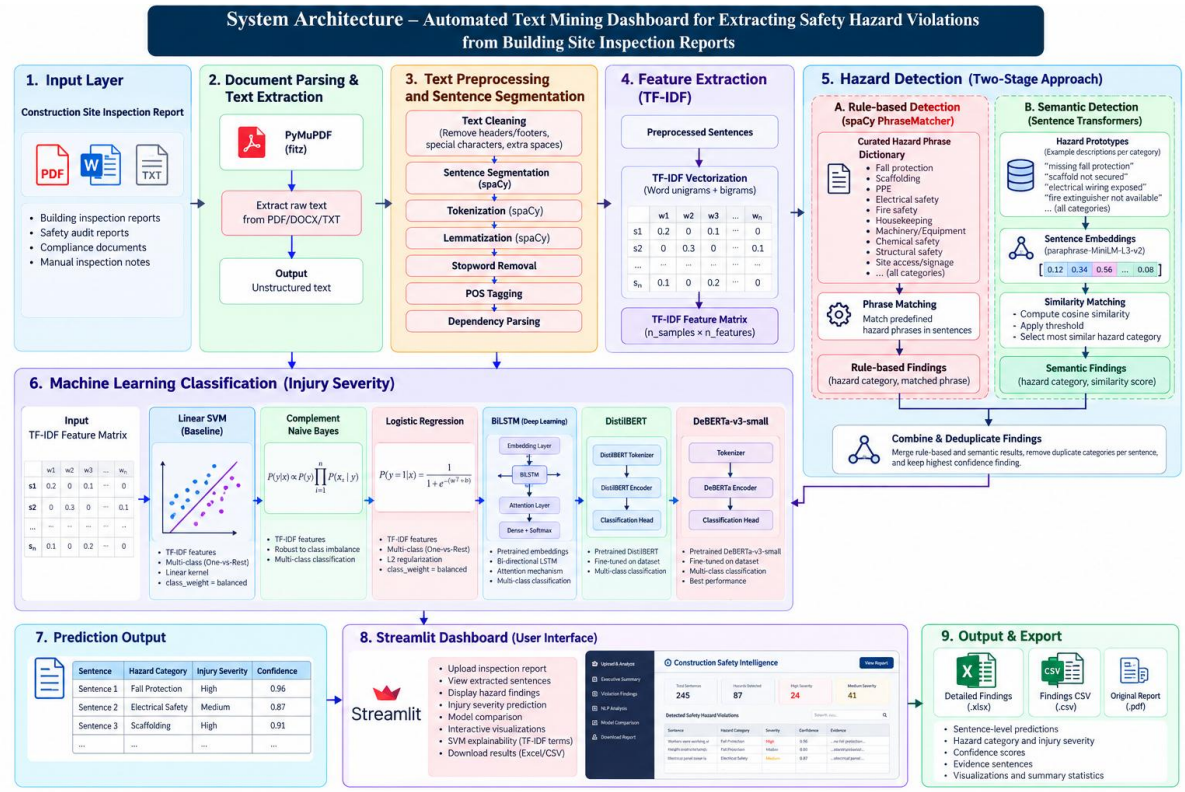


Figure 2. System Architecture.

Module	Processing	Output
Document parser	Extract readable text from PDF/DOCX/TXT.	Raw report text
Text cleaning	Normalize whitespace and remove repeated formatting where reliable.	Clean text
Sentence NLP	Use spaCy sentence boundaries and tokenization.	Sentence list
Rule detector	PhraseMatcher, curated phrases, context/negation checks.	Rule matches with evidence
Semantic detector	Sentence embeddings, prototype similarity, thresholding.	Semantic matches and scores
Classifier	Vectorization/tokenization and inference.	Predicted label; score where supported
Consolidation	Merge duplicate results and preserve provenance.	Structured findings
Dashboard/export	Charts, evidence table, CSV/Excel output.	Reviewable results

4.1.1 Data Description

Working exploration of `Injury Severity.xlsx` indicated approximately 53,550 records and 24 columns. Fields included `abstract`, `event_keyword`, `event_type`, `degree_of_inj_x`, `fatality`, `construction`, `date`, `state_x`, `city_x`, and `sic_code`. Around 10,299 construction-related records were identified in an earlier filtering step. The exact original source name, URL, licence, and filtering code must be verified from the downloaded data before final submission.

A separate cleaned file, `data/clean_training_data.csv`, contained approximately 48,985 rows with `text` and `label`. Observed class counts were approximately: label 0 = 42, label 1 = 19,015, label 2 = 24,078, label 3 = 5,848, and label 4 = 2. The meanings of these numeric labels must be verified from the data dictionary or label-mapping code. Until verified, they should be called class labels rather than assigned unconfirmed severity names. These counts refer to different files/stages and should not be treated as one unchanged dataset.

4.1.2 Data Quality

- Approximately 2,855 exact duplicates were noted in raw spreadsheet exploration.
- One missing abstract was observed in the raw spreadsheet exploration.
- The class distribution is severely imbalanced, especially for the rarest labels.
- Source, licence, inclusion criteria, and preprocessing should be documented.
- Inspection narratives can contain abbreviations, boilerplate, spelling variants, and context-dependent statements.

4.2 Preprocessing

1. Parse the supported document format and preserve the original report for traceability.
2. Normalize whitespace and line breaks; remove headers/footers only when reliably identified.
3. Segment the cleaned text into sentences using spaCy.
4. Retain the original sentence for evidence, even if normalized tokens are used internally.
5. Check labels, missing values, duplicates, and excluded rows.
6. Fit vectorizers only on training data to prevent information leakage.
7. Save preprocessing settings and random seeds so results can be reproduced.

4.2.1 Data Splitting

A stratified split was recorded with seed 42: 35,269 training rows, 3,919 validation rows, and 9,797 test rows. Very rare labels with fewer than three examples were kept only in training; class 4 therefore did not appear in the recorded test set. Generalization performance for that class cannot be inferred from this test set.

4.3 Feature Extraction

4.3.1 TF-IDF

Term Frequency–Inverse Document Frequency (TF-IDF) converts text into weighted sparse features. Terms frequent in a sentence but uncommon across the corpus receive relatively higher weights. Word n-grams can represent short phrases such as “fall protection.” TF-IDF is efficient for high-dimensional text and supports feature-weight interpretation in linear models.

4.3.2 Sentence Embeddings

Sentence-transformer embeddings map sentences to dense vectors. Cosine similarity between a report sentence and a curated hazard prototype can retrieve paraphrases that do not exactly match

a dictionary phrase. Performance depends on the embedding model, prototype quality, and validation-based threshold selection.

4.3.3 Transformer Tokenization

DistilBERT and DeBERTa use subword tokenization and contextual representations. They can model word meaning in context but require more memory and compatible model/tokenizer files and library versions.

4.4 Modelling

4.4.1 Linear SVM

The Linear SVM uses TF-IDF features and a linear decision boundary. It is efficient for sparse, high-dimensional text and its feature weights can support explanation. A decision score is not automatically a calibrated probability; confidence should only be displayed when the implementation supports it appropriately.

4.4.2 Complement Naive Bayes

Complement Naive Bayes is a lightweight text classifier that estimates class evidence using statistics from the complement of each class. It can be useful for imbalanced text classification, though its independence assumptions simplify real language.

4.4.3 BiLSTM with Attention

A Bidirectional Long Short-Term Memory network processes token sequences in both directions. Attention weights can emphasize informative token representations. Performance depends on vocabulary, embeddings, training stability, class imbalance, and the evaluation split.

4.4.4 DistilBERT

DistilBERT is a smaller transformer derived from BERT through knowledge distillation. Fine-tuning adapts contextual representations to the project labels. It offers contextual modelling but costs more computationally than a linear classifier.

4.4.5 DeBERTa-v3-small

DeBERTa-v3-small is a compact transformer using disentangled attention and an enhanced pretraining objective. A locally saved fine-tuned model was used in project experimentation. Local inference requires the correct configuration, tokenizer assets, model weights, and

compatible library versions. Errors should be surfaced rather than silently substituting another model.

Model	Representation	Recorded results / status	Interpretation
Linear SVM	TF-IDF + linear classifier	Common split: 91.65% accuracy; 77.49% macro-F1	Strong practical baseline; supports feature explanations.
Complement Naive Bayes	Sparse text features	Common split: about 82.55% accuracy; 72.64% macro-F1	Lightweight comparison; verify full metric file.
BiLSTM + attention	Sequence model	No final comparable score asserted here	Previous runs differed; evaluate on the common split.
DistilBERT	Fine-tuned transformer	Earlier run: about 91.37% accuracy; 74.53% macro-F1	Verify metrics and foundation.
DeBERTa-v3-small	Fine-tuned transformer	Earlier run: about 91.46% accuracy; 74.65% macro-F1	Verify evaluation setup and saved metrics.

The most defensible directly comparable figures currently recorded are the Linear SVM and Complement Naive Bayes results from the common benchmark split. Transformer and BiLSTM results should be re-evaluated with the same split and label mapping before ranking models.

4.5 Hazard Detection and Finding Consolidation

4.5.1 Rule-Based Phrase Matching

A curated dictionary can cover categories such as fall protection, scaffolding, PPE, electrical safety, fire safety, housekeeping, machinery/equipment, chemical safety, structural safety, and site access/signage. spaCy PhraseMatcher identifies phrase patterns in sentence text. Context checks are needed because a phrase may describe a resolved issue, a negated hazard, or a recommendation rather than an active unsafe condition.

4.5.2 Semantic Detection

The semantic detector compares sentence embeddings with predefined descriptions or prototypes for each hazard category. Similarity is measured and compared with a threshold. Thresholds should be tuned against manually labelled validation examples, not chosen only by intuition.

4.5.3 Consolidation

- Merge rule and semantic results while retaining the detection method.

- Deduplicate repeated category findings for the same sentence where appropriate.
- Preserve original sentence evidence and report context.
- Store scores only when genuinely produced by the relevant model.
- Flag uncertain matches for human review.

5. IMPLEMENTATION AND DASHBOARD

5.1 User Workflow

1. Upload a PDF, DOCX, or TXT inspection report.
2. Extract text and inspect a preview for parsing problems.
3. Clean text and segment it into individual sentences.
4. Run rule-based and semantic hazard detection.
5. Run the selected classifier where the required model is available.
6. Review consolidated findings, evidence, predicted label, and available scores.
7. Export structured findings for downstream review.

5.2 Dashboard Components

Component	Purpose
Upload & Analyze	Accept supported files and run the pipeline.
Executive Summary	Summarize candidate findings and distributions.
Violation Findings	Show hazards with sentence evidence and review flags.
NLP Analysis	Display sentence-level predictions and processing results.
Model Evaluation	Compare models using recorded metrics and reports.
Explainable AI	Show influential TF-IDF terms for an SVM-selected sentence where supported.
Evidence Explorer	Inspect the sentence and its report context.
Export	Download structured results and, where supported, the original report.

5.3 Export Design

An inspection export includes sentence-level predictions, selected model, confidence/similarity score where available, hazard category, evidence, detection method, review flag, corrective-action field and grounded regulator reference where the implementation provides one. Hkj

5.4 Implementation Screenshots

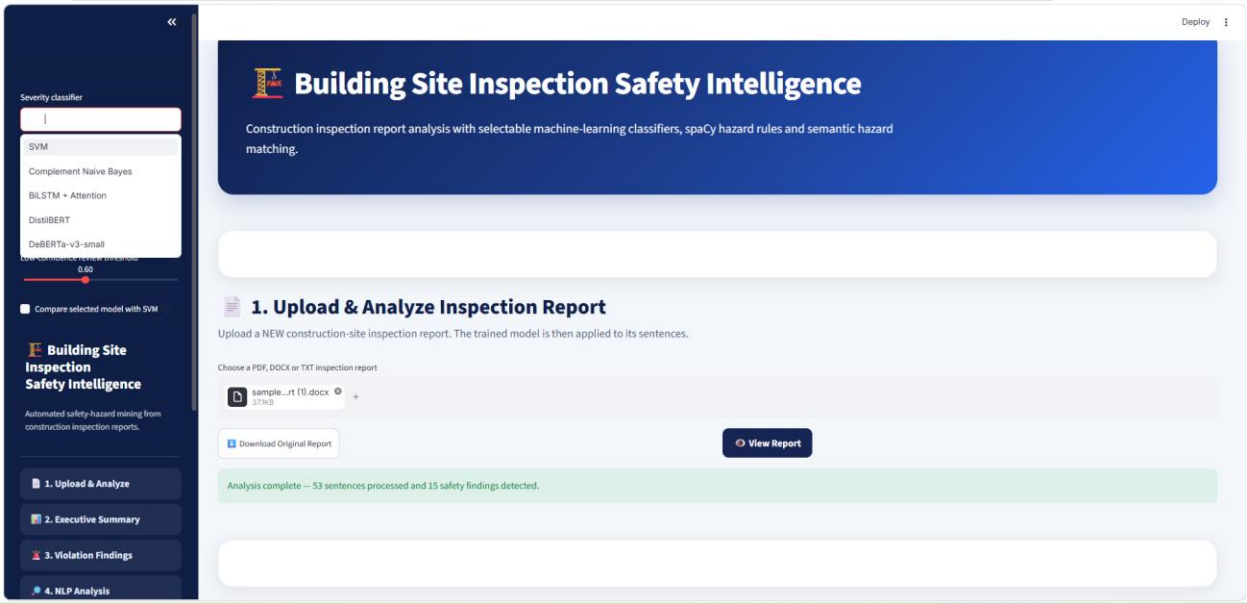
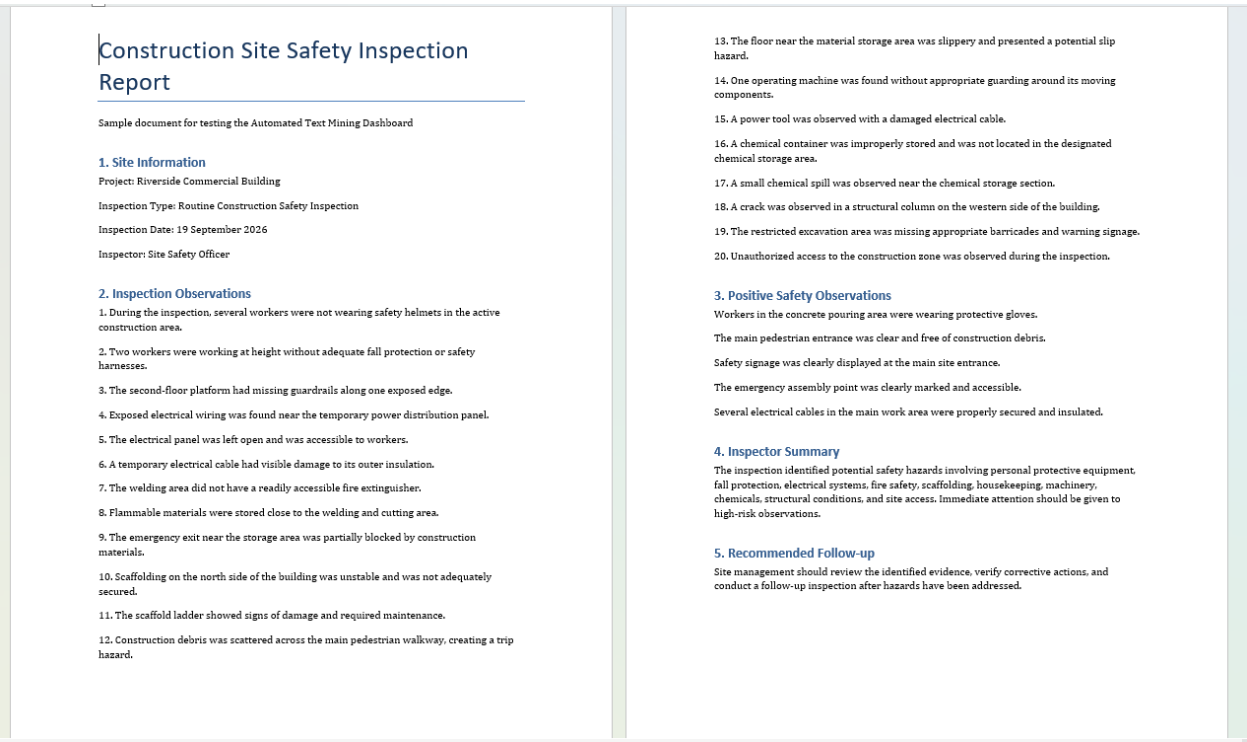


Figure 3. Streamlit landing page and document upload.



Construction Site Safety Inspection Report

Sample document for testing the Automated Text Mining Dashboard

1. Site Information

Project: Riverside Commercial Building

Inspection Type: Routine Construction Safety Inspection

Inspection Date: 19 September 2026

Inspector: Site Safety Officer

2. Inspection Observations

1. During the inspection, several workers were not wearing safety helmets in the active construction area.

2. Two workers were working at height without adequate fall protection or safety harnesses.

3. The second-floor platform had missing guardrails along one exposed edge.

4. Exposed electrical wiring was found near the temporary power distribution panel.

5. The electrical panel was left open and was accessible to workers.

6. A temporary electrical cable had visible damage to its outer insulation.

7. The welding area did not have a readily accessible fire extinguisher.

8. Flammable materials were stored close to the welding and cutting area.

9. The emergency exit near the storage area was partially blocked by construction materials.

10. Scaffolding on the north side of the building was unstable and was not adequately secured.

11. The scaffold ladder showed signs of damage and required maintenance.

12. Construction debris was scattered across the main pedestrian walkway, creating a trip hazard.

13. The floor near the material storage area was slippery and presented a potential slip hazard.

14. One operating machine was found without appropriate guarding around its moving components.

15. A power tool was observed with a damaged electrical cable.

16. A chemical container was improperly stored and was not located in the designated chemical storage area.

17. A small chemical spill was observed near the chemical storage section.

18. A crack was observed in a structural column on the western side of the building.

19. The restricted excavation area was missing appropriate barricades and warning signage.

20. Unauthorized access to the construction zone was observed during the inspection.

3. Positive Safety Observations

Workers in the concrete pouring area were wearing protective gloves.

The main pedestrian entrance was clear and free of construction debris.

Safety signage was clearly displayed at the main site entrance.

The emergency assembly point was clearly marked and accessible.

Several electrical cables in the main work area were properly secured and insulated.

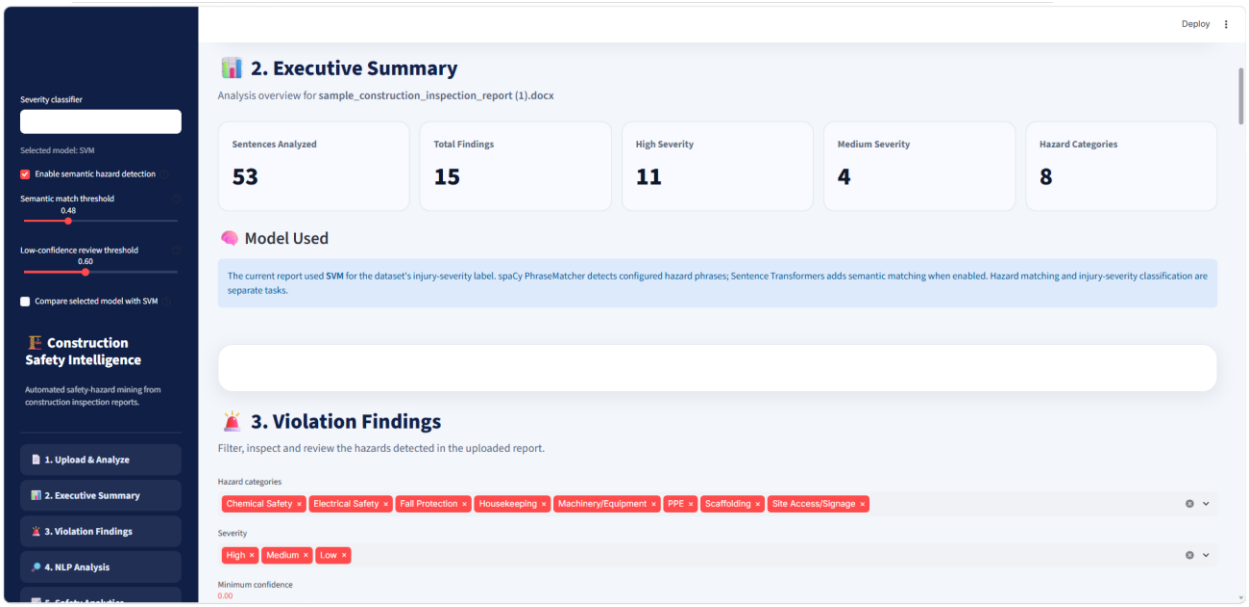
4. Inspector Summary

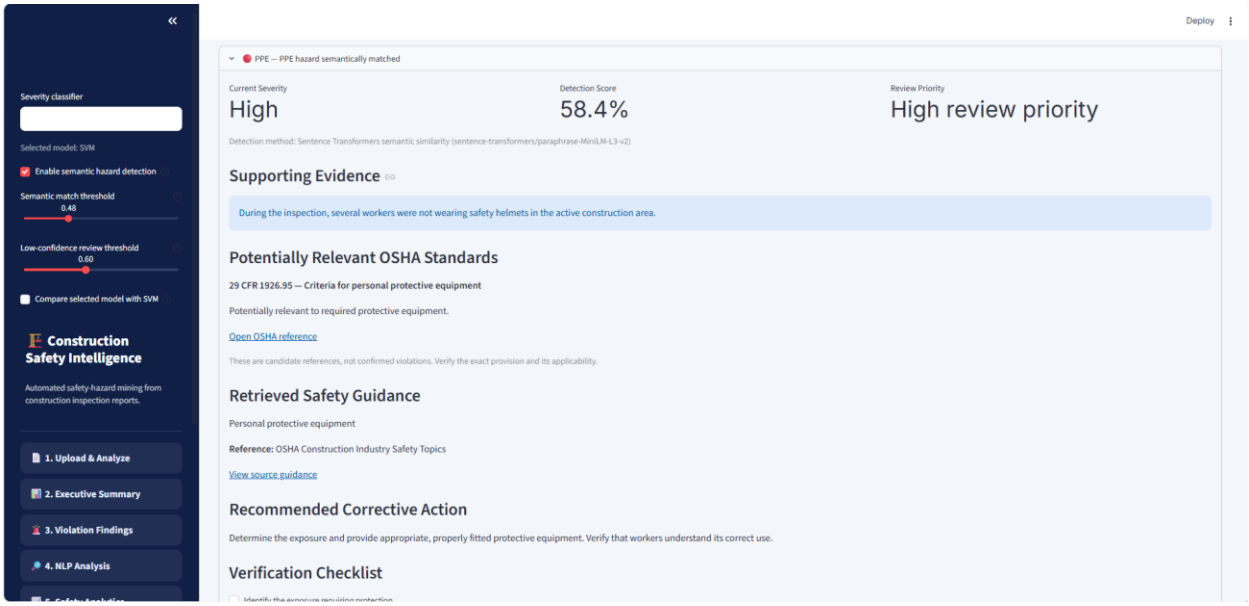
The inspection identified potential safety hazards involving personal protective equipment, fall protection, electrical systems, fire safety, scaffolding, housekeeping, machinery, chemicals, structural conditions, and site access. Immediate attention should be given to high-risk observations.

5. Recommended Follow-up

Site management should review the identified evidence, verify corrective actions, and conduct a follow-up inspection after hazards have been addressed.

Figure 4. Extracted text preview.


Deploy
Severity classifier
Selected model: SVM
Enable semantic hazard detection
Semantic match threshold
0.48
Low-confidence review threshold
0.60
Compare selected model with SVM
Construction Safety Intelligence
Automated safety-hazard mining from construction inspection reports.
1. Upload & Analyze
2. Executive Summary
3. Violation Findings
4. NLP Analysis
2. Executive Summary
Analysis overview for sample_construction_inspection_report (1).docx
Sentences Analyzed
53
Total Findings
15
High Severity
11
Medium Severity
4
Hazard Categories
8
Model Used
The current report used SVM for the dataset's injury-severity label. spaCy PhraseMatcher detects configured hazard phrases; Sentence Transformers adds semantic matching when enabled. Hazard matching and injury-severity classification are separate tasks.
3. Violation Findings
Filter, inspect and review the hazards detected in the uploaded report.
Hazard categories
Chemical Safety
Electrical Safety
Fall Protection
Housekeeping
Machinery/Equipment
PPE
Scaffolding
Site Access/Signage
Severity
High
Medium
Low
Minimum confidence
0.00


Deploy
Severity classifier
Selected model: SVM
Enable semantic hazard detection
Semantic match threshold
0.48
Low-confidence review threshold
0.60
Compare selected model with SVM
Construction Safety Intelligence
Automated safety-hazard mining from construction inspection reports.
1. Upload & Analyze
2. Executive Summary
3. Violation Findings
4. NLP Analysis
PPE — PPE hazard semantically matched
Current Severity
High
Detection Score
58.4%
Review Priority
High review priority
Detection method: Sentence Transformers semantic similarity (sentence-transformers/paraphrase-MiniLM-L3-v2)
Supporting Evidence
During the inspection, several workers were not wearing safety helmets in the active construction area.
Potentially Relevant OSHA Standards
29 CFR 1926.95 — Criteria for personal protective equipment
Potentially relevant to required protective equipment.
Open OSHA reference
These are candidate references, not confirmed violations. Verify the exact provision and its applicability.
Retrieved Safety Guidance
Personal protective equipment
Reference: OSHA Construction Industry Safety Topics
View source guidance
Recommended Corrective Action
Determine the exposure and provide appropriate, properly fitted protective equipment. Verify that workers understand its correct use.
Verification Checklist

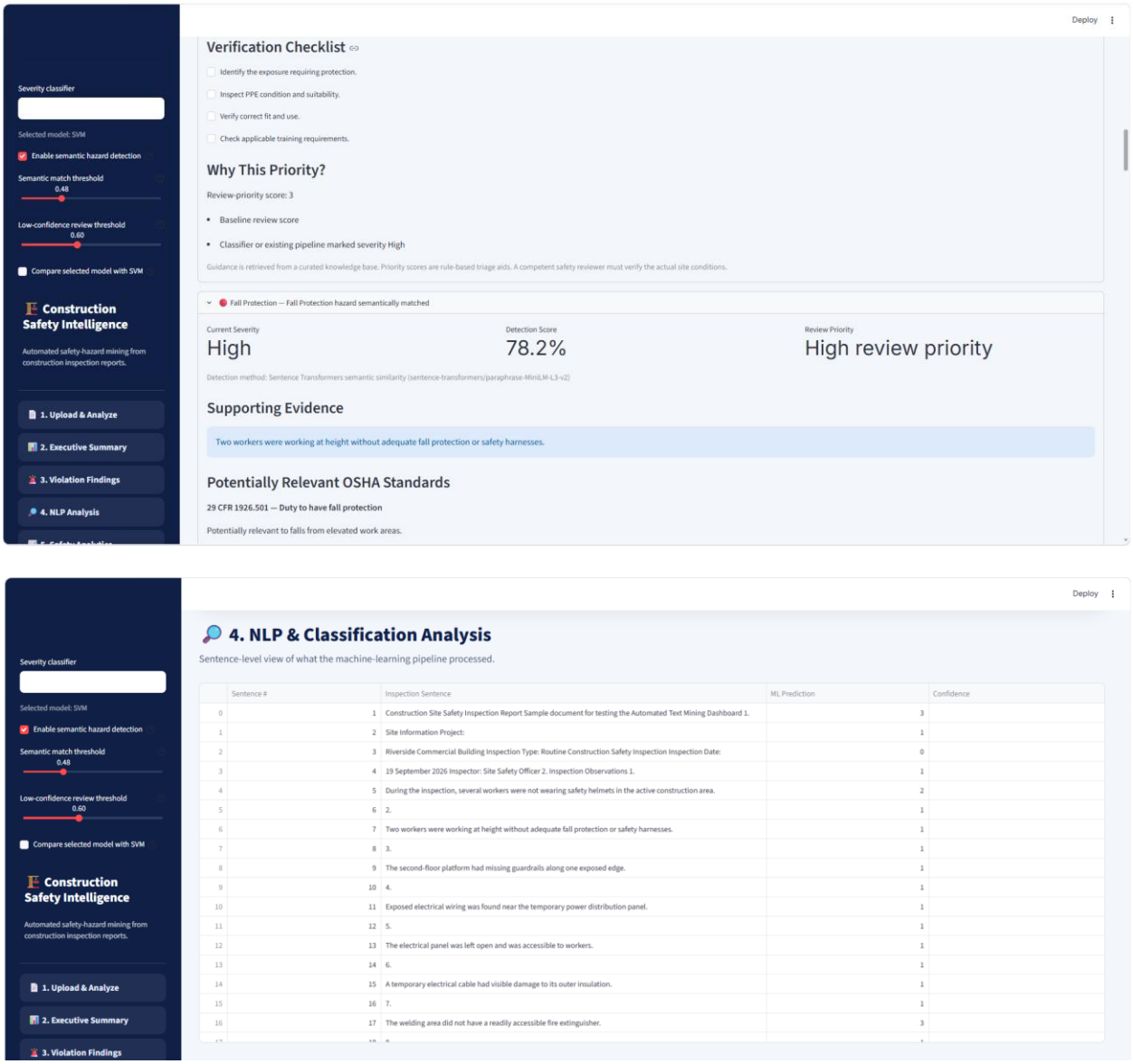


Figure 5. Hazard findings with sentence-level evidence.

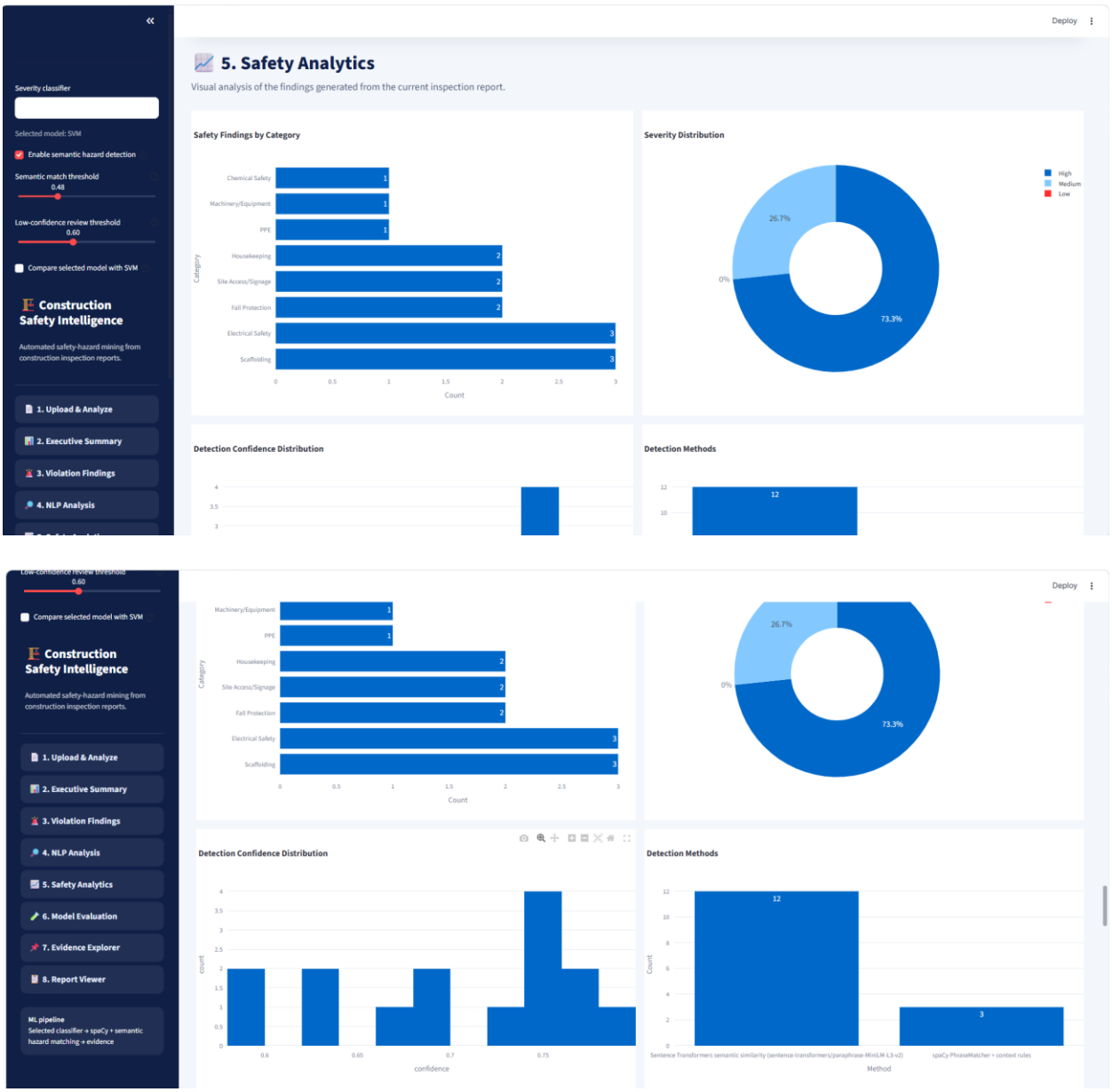
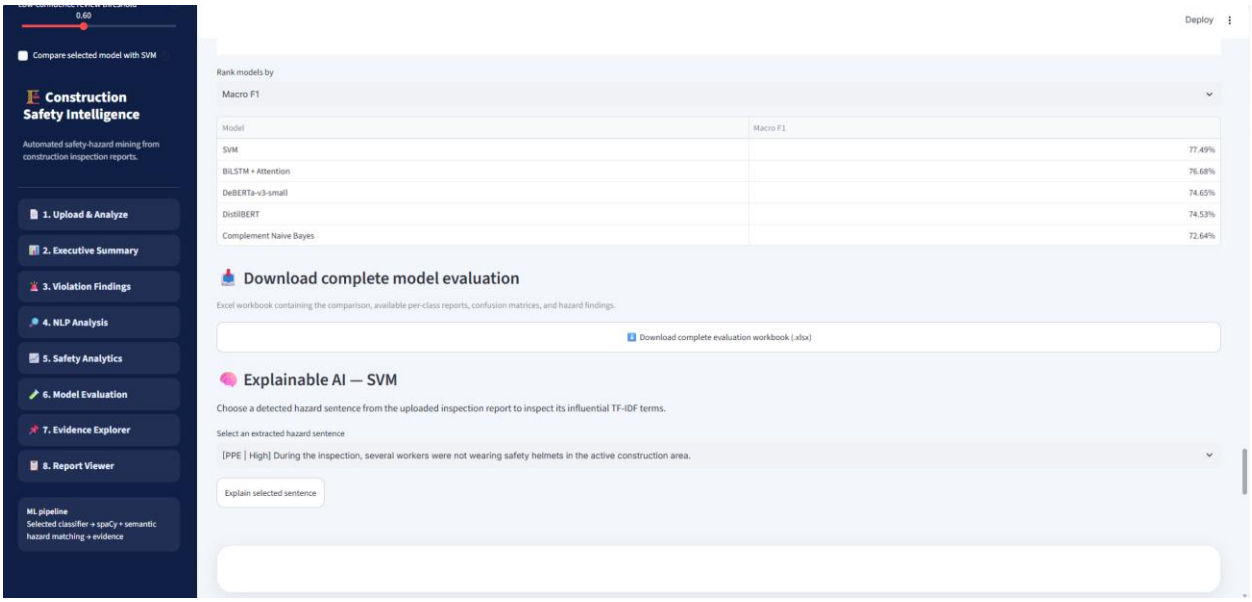


Figure 6. Category and class distribution visualizations.


Low-confidence review threshold
0.60
Compare selected model with SVM
Construction Safety Intelligence
Automated safety-hazard mining from construction inspection reports.
1. Upload & Analyze
2. Executive Summary
3. Violation Findings
4. NLP Analysis
5. Safety Analytics
6. Model Evaluation
7. Evidence Explorer
8. Report Viewer
ML pipeline
Selected classifier → spaCy + semantic hazard matching → evidence
Deploy
6. Model Evaluation
Compare classifier performance and understand the terms influencing SVM predictions.
Model | Accuracy | Macro Precision | Macro Recall | Macro F1
SVM | 91.65% | 84.45% | 73.76% | 77.49%
Complement Naive Bayes | 82.55% | 78.05% | 69.39% | 72.64%
BiLSTM + Attention | 89.87% | 72.05% | 87.32% | 76.68%
DistilBERT | 91.36% | 72.52% | 77.99% | 74.53%
DeBERTa-v3-small | 91.46% | 75.95% | 73.64% | 74.65%
Model performance comparison
Score
Model
Metric
Accuracy
Macro Precision
Macro Recall
Macro F1
SVM
Complement Naive Bayes
BiLSTM + Attention
DistilBERT
DeBERTa-v3-small
Rank models by


Low-confidence review threshold
0.60
Compare selected model with SVM
Construction Safety Intelligence
Automated safety-hazard mining from construction inspection reports.
1. Upload & Analyze
2. Executive Summary
3. Violation Findings
4. NLP Analysis
5. Safety Analytics
6. Model Evaluation
7. Evidence Explorer
8. Report Viewer
ML pipeline
Selected classifier → spaCy + semantic hazard matching → evidence
Deploy
Rank models by
Macro F1
Model | Macro F1
SVM | 77.49%
BiLSTM + Attention | 76.68%
DeBERTa-v3-small | 74.65%
DistilBERT | 74.53%
Complement Naive Bayes | 72.64%
Download complete model evaluation
Excel workbook containing the comparison, available per-class reports, confusion matrices, and hazard findings.
Download complete evaluation workbook (.xlsx)
Explainable AI — SVM
Choose a detected hazard sentence from the uploaded inspection report to inspect its influential TF-IDF terms.
Select an extracted hazard sentence
[PPE | High] During the inspection, several workers were not wearing safety helmets in the active construction area.
Explain selected sentence

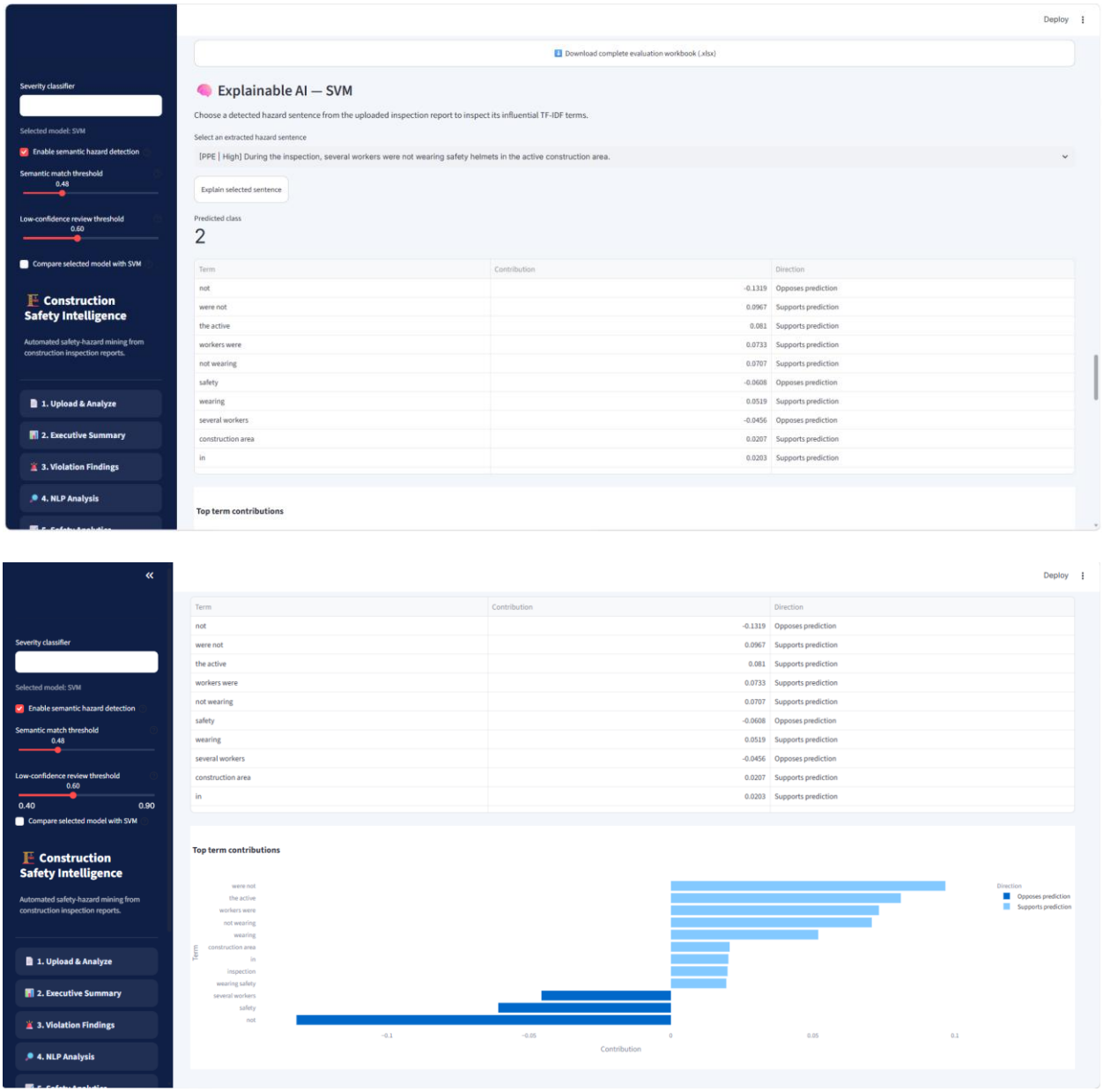
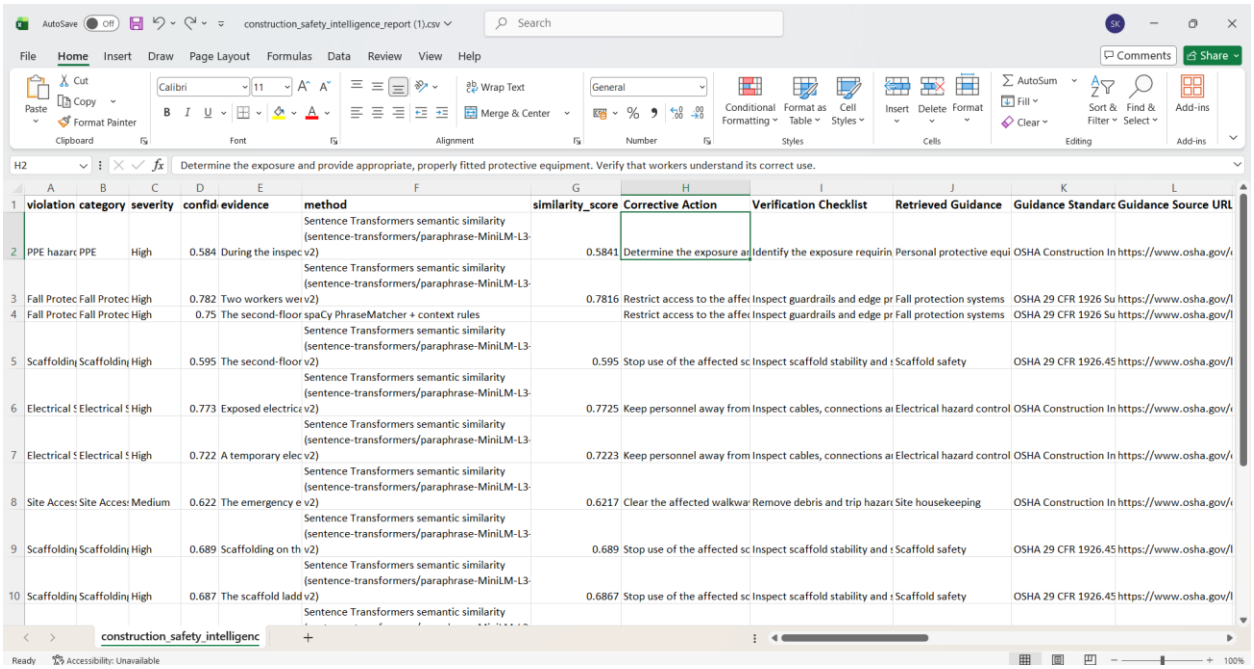
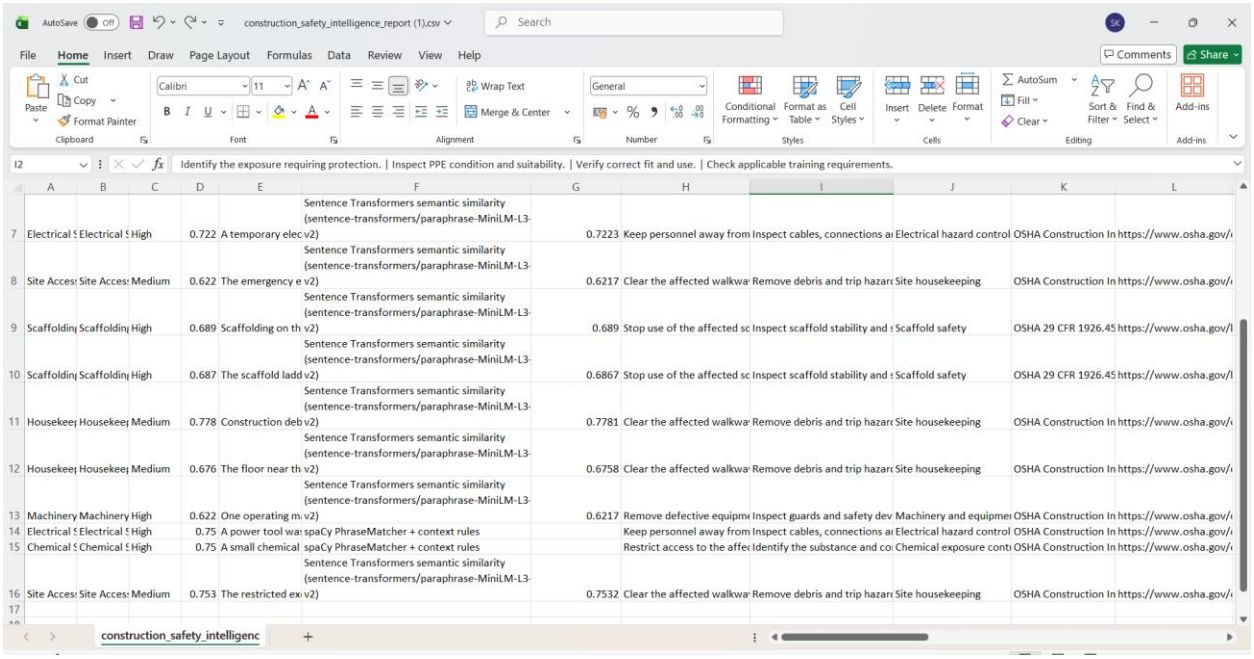


Figure 7. Model evaluation and SVM explainability.



	A	B	C	D	E	F	G	H	I	J	K	L
1	violation	category	severity	confid	evidence	method	similarity_score	Corrective Action	Verification Checklist	Retrieved Guidance	Guidance Standarc	Guidance Source URL
2	PPE hazarc	PPE	High	0.584	During the inspec	Sentence Transformers semantic similarity (sentence-transformers/paraphrase-MiniLM-L3-v2)	0.5841	Determine the exposure a	Identify the exposure requirin	Personal protective equi	OSHA Construction In	https://www.osha.gov/
3	Fall Protec	Fall Protec	High	0.782	Two workers wer	Sentence Transformers semantic similarity (sentence-transformers/paraphrase-MiniLM-L3-v2)	0.7816	Restrict access to the affe	Inspect guardrails and edge pr	Fall protection systems	OSHA 29 CFR 1926 Su	https://www.osha.gov/l
4	Fall Protec	Fall Protec	High	0.75	The second-floor	spaCy PhraseMatcher + context rules		Restrict access to the affe	Inspect guardrails and edge pr	Fall protection systems	OSHA 29 CFR 1926 Su	https://www.osha.gov/l
5	Scaffoldin	Scaffoldin	High	0.595	The second-floor	Sentence Transformers semantic similarity (sentence-transformers/paraphrase-MiniLM-L3-v2)	0.595	Stop use of the affected sc	Inspect scaffold stability and s	Scaffold safety	OSHA 29 CFR 1926.45	https://www.osha.gov/l
6	Electrical S	Electrical S	High	0.773	Exposed electric	Sentence Transformers semantic similarity (sentence-transformers/paraphrase-MiniLM-L3-v2)	0.7725	Keep personnel away from	Inspect cables, connections a	Electrical hazard control	OSHA Construction In	https://www.osha.gov/
7	Electrical S	Electrical S	High	0.722	A temporary elec	Sentence Transformers semantic similarity (sentence-transformers/paraphrase-MiniLM-L3-v2)	0.7223	Keep personnel away from	Inspect cables, connections a	Electrical hazard control	OSHA Construction In	https://www.osha.gov/
8	Site Acces	Site Acces	Medium	0.622	The emergency e	Sentence Transformers semantic similarity (sentence-transformers/paraphrase-MiniLM-L3-v2)	0.6217	Clear the affected walkwa	Remove debris and trip hazarc	Site housekeeping	OSHA Construction In	https://www.osha.gov/
9	Scaffoldin	Scaffoldin	High	0.689	Scaffolding on th	Sentence Transformers semantic similarity (sentence-transformers/paraphrase-MiniLM-L3-v2)	0.689	Stop use of the affected sc	Inspect scaffold stability and s	Scaffold safety	OSHA 29 CFR 1926.45	https://www.osha.gov/l
10	Scaffoldin	Scaffoldin	High	0.687	The scaffold ladd	Sentence Transformers semantic similarity (sentence-transformers/paraphrase-MiniLM-L3-v2)	0.6867	Stop use of the affected sc	Inspect scaffold stability and s	Scaffold safety	OSHA 29 CFR 1926.45	https://www.osha.gov/l



	A	B	C	D	E	F	G	H	I	J	K	L
7	Electrical S	Electrical S	High	0.722	A temporary elec	Sentence Transformers semantic similarity (sentence-transformers/paraphrase-MiniLM-L3-v2)	0.7223	Keep personnel away from	Inspect cables, connections a	Electrical hazard control	OSHA Construction In	https://www.osha.gov/
8	Site Acces	Site Acces	Medium	0.622	The emergency e	Sentence Transformers semantic similarity (sentence-transformers/paraphrase-MiniLM-L3-v2)	0.6217	Clear the affected walkwa	Remove debris and trip hazarc	Site housekeeping	OSHA Construction In	https://www.osha.gov/
9	Scaffoldin	Scaffoldin	High	0.689	Scaffolding on th	Sentence Transformers semantic similarity (sentence-transformers/paraphrase-MiniLM-L3-v2)	0.689	Stop use of the affected sc	Inspect scaffold stability and s	Scaffold safety	OSHA 29 CFR 1926.45	https://www.osha.gov/l
10	Scaffoldin	Scaffoldin	High	0.687	The scaffold ladd	Sentence Transformers semantic similarity (sentence-transformers/paraphrase-MiniLM-L3-v2)	0.6867	Stop use of the affected sc	Inspect scaffold stability and s	Scaffold safety	OSHA 29 CFR 1926.45	https://www.osha.gov/l
11	Housekeer	Housekeer	Medium	0.778	Construction deb	Sentence Transformers semantic similarity (sentence-transformers/paraphrase-MiniLM-L3-v2)	0.7781	Clear the affected walkwa	Remove debris and trip hazarc	Site housekeeping	OSHA Construction In	https://www.osha.gov/
12	Housekeer	Housekeer	Medium	0.676	The floor near th	Sentence Transformers semantic similarity (sentence-transformers/paraphrase-MiniLM-L3-v2)	0.6758	Clear the affected walkwa	Remove debris and trip hazarc	Site housekeeping	OSHA Construction In	https://www.osha.gov/
13	Machinery	Machinery	High	0.622	One operating m	Sentence Transformers semantic similarity (sentence-transformers/paraphrase-MiniLM-L3-v2)	0.6217	Remove defective equipme	Inspect guards and safety dev	Machinery and equipmer	OSHA Construction In	https://www.osha.gov/
14	Electrical S	Electrical S	High	0.75	A power tool wa	spaCy PhraseMatcher + context rules		Keep personnel away from	Inspect cables, connections a	Electrical hazard control	OSHA Construction In	https://www.osha.gov/
15	Chemical S	Chemical S	High	0.75	A small chemical	spaCy PhraseMatcher + context rules		Restrict access to the affe	Identify the substance and co	Chemical exposure contr	OSHA Construction In	https://www.osha.gov/
16	Site Acces	Site Acces	Medium	0.753	The restricted ex	Sentence Transformers semantic similarity (sentence-transformers/paraphrase-MiniLM-L3-v2)	0.7532	Clear the affected walkwa	Remove debris and trip hazarc	Site housekeeping	OSHA Construction In	https://www.osha.gov/
17												

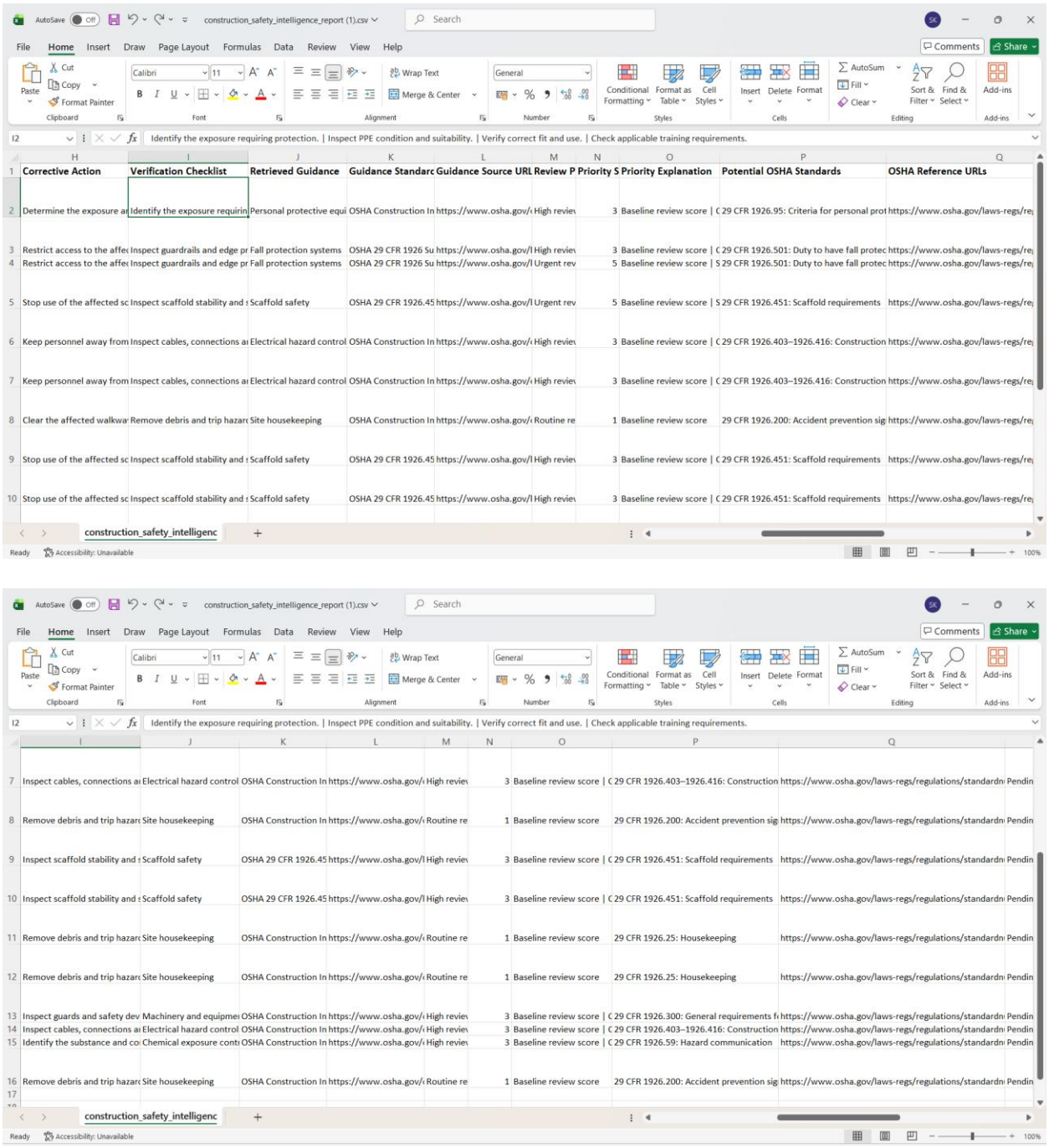


Figure 8. Exported findings workbook or CSV.

6. EVALUATION AND DISCUSSION

6.1 Evaluation Metrics

Accuracy is the fraction of all predictions that are correct. Precision is the fraction of predicted class members that are correct; recall is the fraction of actual class members retrieved; F1 is the harmonic mean of precision and recall. Macro averages weight each class equally and are useful under imbalance. The final evaluation should include per-class precision, recall, F1, support, confusion matrix, and test class distribution.

6.2 Preliminary Recorded Results

Model	Accuracy	Macro precision	Macro recall	Macro-F1
Linear SVM (common split)	91.65%	84.45%	73.76%	77.49%
Complement Naive Bayes (common split)	~82.55%	Verify final metrics	Verify final metrics	~72.64%
DistilBERT (earlier run)	~91.37%	~72.52%	~77.99%	~74.53%
DeBERTa-v3-small (earlier run)	~91.46%	~75.95%	~73.64%	~74.65%

These are preliminary recorded project results. The Complement Naive Bayes precision and recall should be copied from the final benchmark metrics file because earlier summaries differed slightly. Transformer scores must be checked for split consistency before direct comparison.

6.3 Interpretation

The common-split figures suggest that Linear SVM is a strong practical baseline for the prepared data. Its macro-F1 is more informative than accuracy alone under severe imbalance. Complement Naive Bayes provides a lightweight alternative with lower recorded scores. Transformer results may be useful but require a common evaluation setup before comparative claims are justified.

6.4 Limitations and Responsible Use

- Severe class imbalance makes rare-class metrics unstable.
- A rare class was absent from the recorded test set, preventing a test estimate for that class.
- Numeric labels must be mapped to documented meanings before calling them severity levels.
- The raw spreadsheet and cleaned CSV represent different processing stages.

- Phrase matching may miss unfamiliar wording and can produce false positives.
- Semantic similarity depends on prototypes and threshold calibration.
- Scores from different model families may not be calibrated or directly comparable.
- Transformer inference can fail because of model path or dependency mismatches.
- Safety findings should be reviewed by qualified personnel; the dashboard is not a compliance certification system.

7. CONCLUSION AND FUTURE WORK

This project presents a text-mining workflow for finding candidate safety hazards in building-site inspection reports. It combines document parsing, spaCy sentence processing, rule-based phrase matching, semantic similarity, supervised classification, and a Streamlit dashboard that displays findings with supporting evidence. The dashboard is intended to make narrative information easier to explore and export while preserving human review.

The recorded common-split experiment indicates that TF-IDF with Linear SVM is a strong baseline, with 91.65% accuracy and 77.49% macro-F1. Results must be interpreted in light of class imbalance, rare-class limitations, and the need to verify label meanings and dataset provenance. All models should be evaluated on the same split before making direct comparative claims.

The system integrates machine learning and natural language processing to support construction safety analysis. By combining classification, semantic matching, and evidence extraction, it provides a structured approach to identifying potential hazards. The dashboard improves accessibility, supports data-driven review, and helps safety personnel examine findings more efficiently while maintaining human oversight.

7.1 Future Work

- Verify and cite the exact source, licence, data dictionary, and filtering procedure.
- Regenerate all metrics from a single versioned split and include per-class reports/confusion matrices.
- Improve rare-class evaluation with additional verified labelled examples.
- Tune semantic thresholds using manually annotated inspection sentences.
- Improve negation, temporal, and corrective-action context handling.

- Add reviewer feedback to improve the phrase dictionary and labels.
- Validate hazard precision and recall with safety professionals.
- Improve reproducibility, model-loading diagnostics, and secure deployment.

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